

Solution Requirements

Date	30 June 2026
Team ID	N/A
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the application through the Streamlit interface. No authentication or login module is implemented in the current version.	Medium
2	Authorization levels	All users have equal access to generate conversation starters, self-introductions, networking tips, perform fact verification, and view conversation history.	Medium

3	External interfaces	Integrates with the Groq API for AI-generated responses and the Wikipedia API for fact verification	High
4	Transactions processing	Accepts user details such as interests, background, profession, and event context, processes requests through the FastAPI backend, generates AI responses, verifies facts, and stores conversation history in SQLite.	High
5	Reporting	Displays AI-generated conversation starters, personalized self-introductions, networking tips, verified facts, and previous conversation history.	High
6	Business rules, etc.	User interests and event details are required before generating networking suggestions. Conversation history is stored locally in the SQLite (networking.db) database.	High
7	Compliance to laws or regulations	Uses publicly available information from Wikipedia and avoids storing sensitive personal information beyond conversation history required for application functionality.	Medium
8	<i>Other</i>	Users can retrieve previous conversations from the SQLite database to improve networking preparation.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate AI responses quickly for a smooth user experience.	Response within 3–5 seconds
2	Scalability	Modular architecture allows future AI models and APIs to be added easily.	Supports additional modules with minimal code changes
3	Security & Data Privacy	Input validation using FastAPI and Pydantic. Conversation history is stored locally in SQLite. API keys are managed through environment variables (.env).	Secure API validation with no sensitive data exposure
4	Reliability & Availability	The application should remain stable during continuous usage and handle API requests reliably.	Approximately 99% successful request processing under normal conditions.
5	Usability & Accessibility	Simple and intuitive Streamlit interface suitable for students, professionals, and first-time users.	Easy navigation with minimal user training.
6	Other	Modular FastAPI backend, reusable service modules, and organized project structure for easier maintenance and future enhancements.	Easy maintenance and future enhancements